Plurality Collapse: Measuring the Dimensionality Gap Between Human and LLM Moral Reasoning

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Abstract

Large language models are increasingly used to simulate human moral judgment, yet it is unclear whether they capture the full diversity of human moral reasoning or collapse it into a narrower subspace. We embed human and LLM rationales from 10,826 everyday moral dilemmas into the Kaleido-XL moral knowledge encoder and measure the geometric dimensionality of each source's embedding distribution. Human rationales require 448 PCA components for 90% variance explanation compared to 317–332 for seven LLMs at matched sample sizes—a 28–41% dimensionality advantage. Cross-space projection confirms the asymmetry is directional: human principal components capture 87.3% of LLM variance, but LLM components capture only 81.6% of human variance. The unexplained human variance encodes rare moral values and situational concreteness (70%) alongside informal linguistic register (30%). The dimensionality gap widens on high-consensus dilemmas and LLMs maintain fixed inter-model convergence regardless of how controversial a dilemma is among humans.

1 Question

Do large language models flatten the diversity of human moral reasoning? When LLMs generate rationales for everyday moral dilemmas, do they capture the same breadth of moral perspectives that humans express, or do they collapse a high-dimensional space of values, framings, and situational judgments into a lower-dimensional subspace?

This question matters because LLMs are increasingly deployed in contexts that require moral reasoning—content moderation, advisory systems, ethical review tools. If LLMs systematically underrepresent certain moral perspectives, their outputs will reflect a narrower value system than the populations they serve. The concern is not that LLMs produce *wrong* moral judgments, but that they produce *insufficiently diverse* ones, missing the plurality of legitimate moral viewpoints that characterizes genuine human discourse.

We operationalize "diversity of moral reasoning" as the geometric dimensionality of rationale embeddings in a moral knowledge representation space. If LLM rationales occupy fewer effective dimensions than human rationales, that constitutes measurable evidence of plurality collapse.

2 Do (Methods)

2.1 Dataset

We use the Sachdeva et al. *Normative Evaluation of LLMs on Everyday Dilemmas* dataset (`ucberkeley-dlab`), which contains 10,826 moral dilemmas from Reddit's "Am I the Asshole" forum. Each dilemma includes the original post, a top human comment (rationale), and moral rationales generated by seven LLMs: GPT-3.5, GPT-4, Claude, Bison, Gemma, Mistral, and

Llama. Each LLM provides 2–3 rationales per dilemma (yielding 21,652–32,478 LLM rationales per model). The dataset also provides weighted community agreement scores across four verdict categories (NTA, YTA, ESH, NAH).

2.2 Embedding Extraction

We encode all rationales using the encoder of Kaleido-XL, an encoder-decoder model trained on the ValuePrism dataset to map natural language descriptions of actions to moral value expressions. Each rationale is formatted with Kaleido's generation template (`[Generate]:\tAction: TEXT`) and passed through the encoder. The final hidden states are mean-pooled across token positions, producing a 2048-dimensional embedding per rationale. This encoder was chosen because its representation space is explicitly structured around moral knowledge, making dimensionality in this space a meaningful proxy for moral reasoning diversity.

2.3 Dimensionality Analysis

For each source (human, each LLM, and all LLMs concatenated), we fit PCA on the full embedding matrix and compute:

- **Participation ratio** (PR): $\mathrm{PR} = (\sum \lambda_i)^2 / \sum \lambda_i^2$, measuring the effective number of dimensions.
- **Components for 90%/95% variance**: the number of principal components k such that cumulative explained variance ≥ 0.90 or 0.95.

2.4 Cross-Space Projection

To test whether one source's reasoning subspace contains another's, we fit PCA on source A, project source B's data onto A's top-k components (where k is A's 90% threshold), and measure the fraction of B's variance captured. If human components explain LLM variance better than LLM components explain human variance, LLMs occupy a subspace of the human moral value space.

2.5 Unexplained Variance Inspection

For rationales with high reconstruction error (poorly captured by the other space's PCA basis), we:

1. Run Kaleido-XL's decoder (`model.generate()`) to produce moral value expressions.
2. Compare value diversity between poorly-captured and well-captured groups.
3. Apply TF-IDF analysis to identify distinguishing textual features.
4. Categorize distinguishing terms as stylistic (Reddit slang, profanity) or content (moral concepts, situational nouns).

2.6 Consensus Stratification

We bucket dilemmas by moral consensus level (maximum weighted agreement across verdicts): high (> 0.8), medium (0.5–0.8), and low (< 0.5). Within each bucket, we subsample each non-human source to match the human embedding count (with 5 random seeds) and recompute PCA, controlling for sample size confounds. We also compute per-dilemma LLM convergence as the average pairwise cosine distance among all LLM embeddings for each dilemma.

2.7 Robustness Controls

- **Sample size**: All LLM sources subsampled to $n = 10{,}826$ (matching human) before comparison.

- **Text length**: Pearson and Spearman correlation between average rationale word count and PR.
- **Nonlinearity**: Kernel PCA (RBF kernel) as a check on linear PCA assumptions.

3 Author's Rationale

Prior work on LLM alignment and bias has largely focused on output-level metrics: agreement with human labels, toxicity scores, or demographic fairness. These approaches measure whether LLMs produce acceptable outputs but cannot assess whether they capture the *full distribution* of human moral perspectives. An LLM could achieve high label agreement while systematically underrepresenting minority viewpoints.

We measure diversity at the representation level rather than the output level. By embedding rationales into a moral knowledge space (Kaleido-XL) and analyzing the geometry of the resulting distributions, we can directly quantify how many independent dimensions of moral variation each source uses. This approach has several advantages:

1. It is **model-agnostic**: any text-based rationale can be embedded and compared.
2. It captures **implicit** moral content, not just explicit value labels.
3. It provides a **continuous** measure of diversity rather than a binary agree/disagree metric.
4. PCA dimensionality is **well-understood** statistically, with clear robustness checks (subsampling, kernel PCA, participation ratio vs. component counts).

The cross-space projection analysis goes beyond measuring dimensionality to testing *containment*: whether one source's moral reasoning lives inside another's. This directional test is the strongest form of evidence for plurality collapse—it shows not just that LLMs use fewer dimensions, but that the dimensions they use are a proper subset of the human dimensions.

4 Findings

4.1 Human Moral Reasoning Is Higher-Dimensional

Table 1 presents the core dimensionality metrics. Human rationales require 448 components for 90% variance explanation, compared to 329–341 for individual LLMs and 347 for the combined LLM pool. The participation ratio is less sensitive to this gap (human PR = 52.40 vs. LLM range 42.65–54.36), as it is dominated by the top eigenvalues. The components-for-90% metric captures the "long tail" where human reasoning extends into dimensions LLMs do not use.

Table 1: PCA dimensionality metrics per source (full sample sizes).

Source	n	PR	Kernel PR	Comp. 90%	Comp. 95%
Human	10,826	52.40	52.92	**448**	**785**
GPT-3.5	32,478	49.76	52.41	329	642
GPT-4	21,652	54.36	56.70	341	658
Claude	32,478	52.06	54.17	340	653
Bison	32,478	52.00	53.78	336	652
Gemma	32,478	46.16	48.62	334	653
Mistral	32,455	42.65	45.82	332	654
Llama	32,478	51.08	53.25	333	651
All LLM	216,497	51.95	54.19	347	679

4.1.1 Robustness: Sample Size

Subsampling all sources to $n = 10{,}826$ with 5 seeds confirms the gap is not a sample size artifact. Human retains 448 components for 90% variance while subsampled LLMs range from 317.0 (GPT-3.5) to 332.2 (GPT-4), with standard deviations below 1 component (Figure 1).

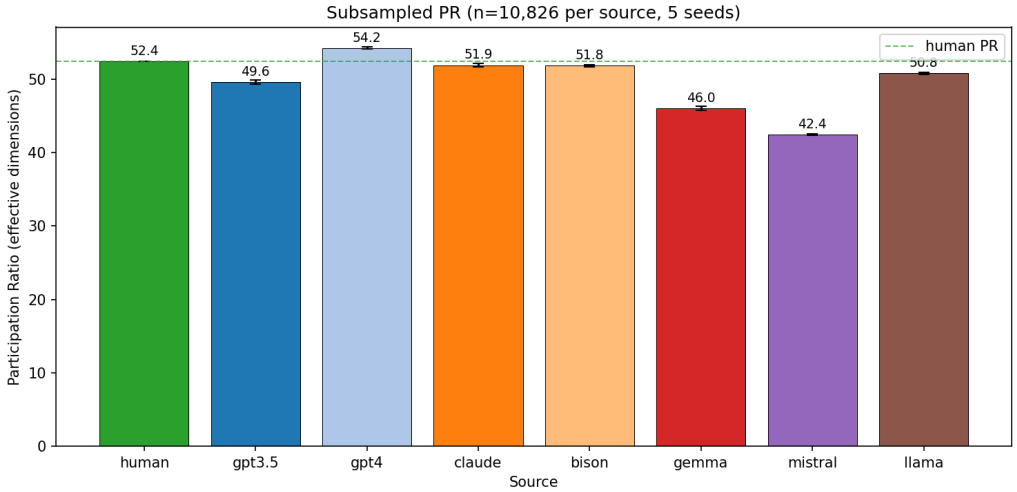

Figure 1: Participation ratio at matched sample sizes ($n = 10{,}826$, 5 seeds). Error bars show standard deviation across seeds.

4.1.2 Robustness: Text Length

There is no significant correlation between average rationale length and participation ratio (Pearson $r = -0.092$, $p = 0.83$; Spearman $\rho = -0.524$, $p = 0.18$; $n = 8$ sources). Human rationales are the shortest (57.2 words on average) yet among the highest-dimensional, making any length-driven confound work against our finding.

4.2 LLMs Occupy a Subspace of Human Moral Reasoning

Cross-space projection reveals a directional asymmetry (Figure 2). Human PCs ($k = 448$) capture 87.3% of all-LLM variance, but all-LLM PCs ($k = 347$) capture only 81.6% of human variance—an asymmetry of +5.7 percentage points. Even at matched $k = 347$, human PCs capture 84.5% of LLM variance compared to 81.6% in the reverse direction (+2.9 pp at equal k).

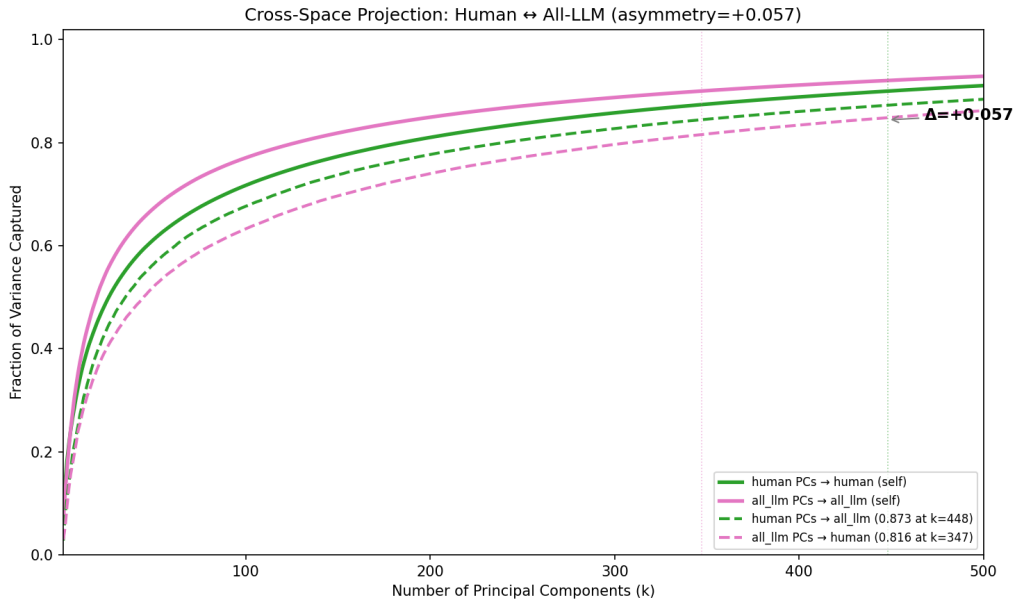

Figure 2: Variance captured as a function of k principal components. Solid lines: self-projection. Dashed lines: cross-projection. The gap between the two dashed lines quantifies the asymmetry.

The full cross-projection heatmap (Figure 3) shows that every individual LLM's PCs capture only 77.5–81.6% of human variance, while human PCs capture 85.8–88.6% of each LLM.

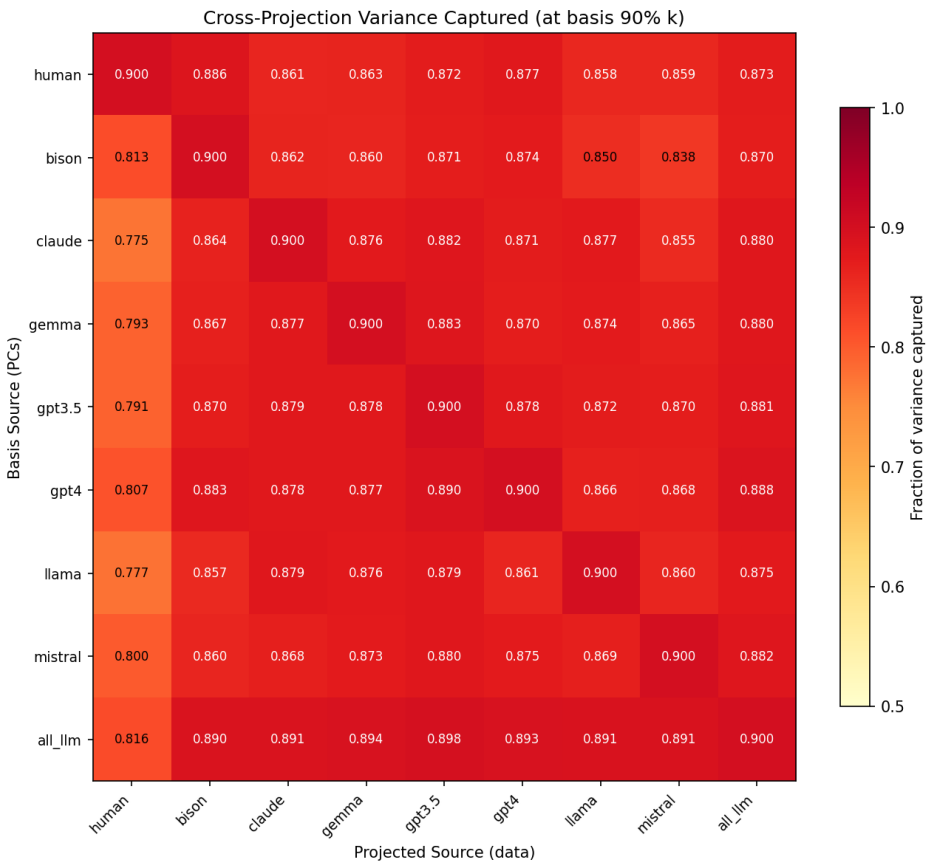

Figure 3: Variance captured at each basis source's 90% k for all directed pairs. Rows = basis (PCs), columns = projected data. Off-diagonal values below the diagonal (LLM→human) are systematically lower than above (human→LLM).

4.3 Decomposing the Unexplained Variance

4.3.1 What LLMs Miss About Humans (18.4%)

Poorly-captured human rationales (high reconstruction error under LLM PCs) invoke 58 unique Kaleido values compared to 28 for well-captured rationales—a 2.1× diversity advantage (Figure 4). Well-captured rationales are dominated by Respect (29.0%) and Family harmony (23.5%); poorly-captured rationales spread across Freedom of speech (6.5%), Safety (6.0%), Health (4.5%), and 55 other values.

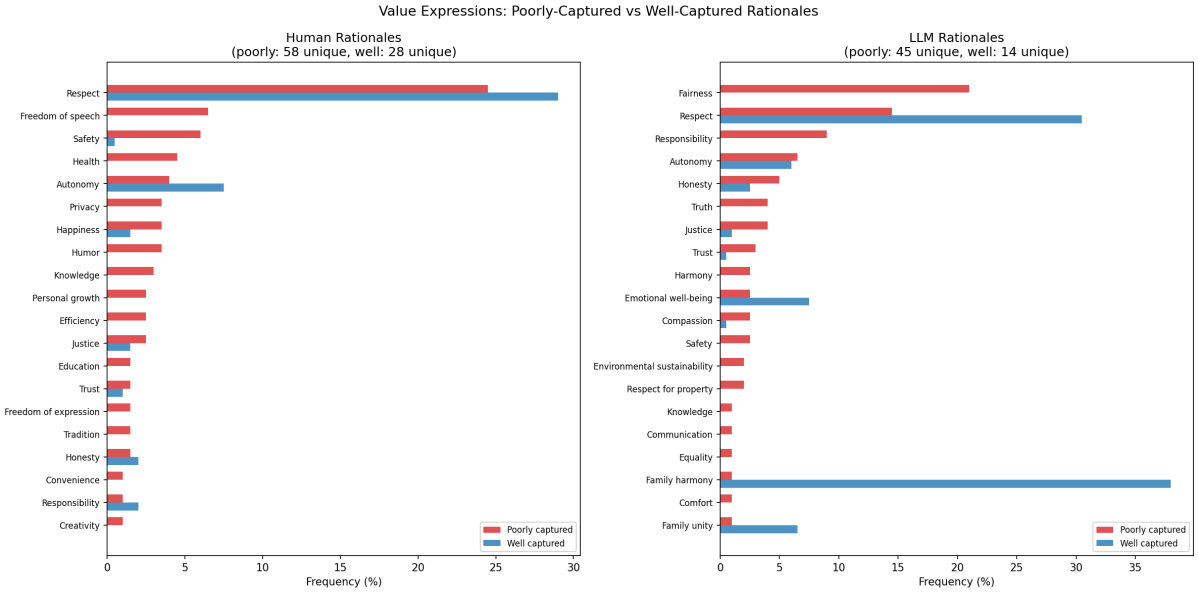

Figure 4: Value expression distributions for poorly-captured vs. well-captured rationales. Left: human rationales projected onto LLM PCs. Right: LLM rationales projected onto human PCs.

TF-IDF analysis of the poorly-captured human rationales reveals a mixed signal: 70% of the top 30 distinguishing terms are content-related (concrete nouns like *lawyer*, *door*, *clothes*, *service*) while 30% are stylistic markers (*nta*, *lol*, *wtf*, *fuck*, *lmao*). See Figure 5 and Table 2.

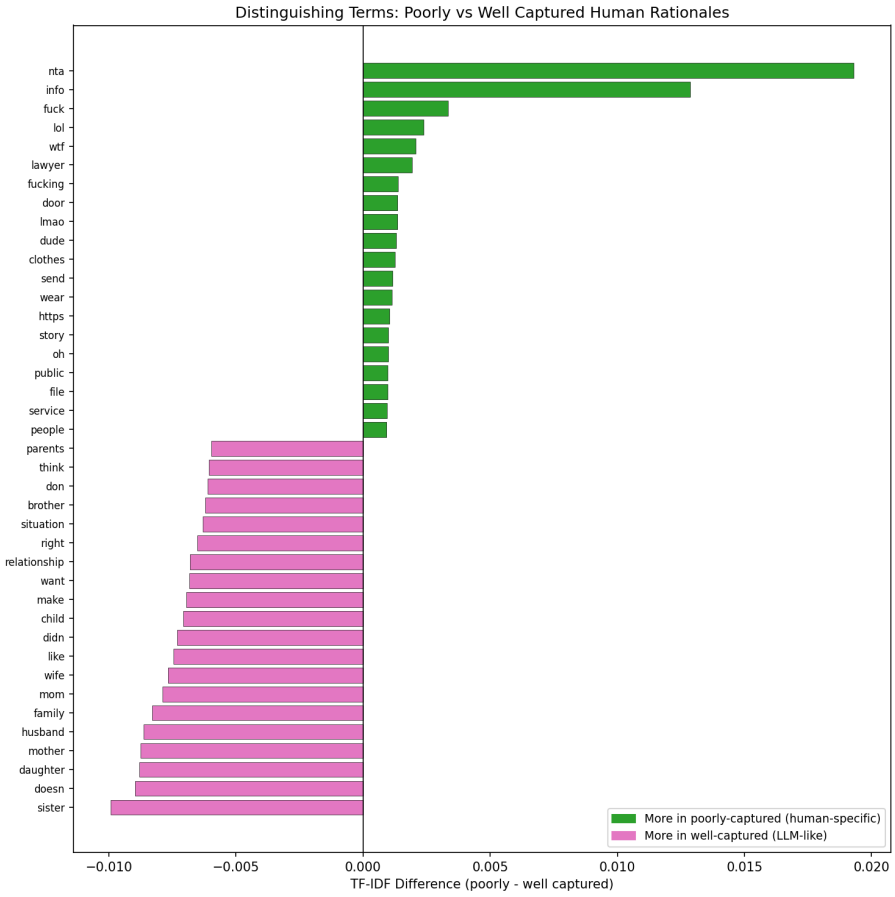

Figure 5: TF-IDF terms distinguishing poorly-captured (green, human-specific) from well-captured (pink, LLM-like) human rationales.

Table 2: Categorization of top 30 TF-IDF terms for poorly-captured rationales.

Side	Stylistic	Content	% Stylistic
Human poorly-captured	9	21	30.0%
LLM poorly-captured	1	29	3.3%

4.3.2 What Humans Miss About LLMs (12.7%)

LLM-specific variance is almost entirely content-driven (96.7% of distinguishing terms). The top TF-IDF terms are analytical (*information*, *determine*, *judgment*, *definitive*, *context*, *assessment*), reflecting a distinctive hedging register LLMs use for ambiguous moral cases.

4.3.3 Value Overlap

Jaccard similarity between human-unexplained and LLM-unexplained value sets is 0.157 (Table 3), confirming these are genuinely different moral dimensions. Human-specific values include Freedom of speech, Cultural diversity, and Humor; LLM-specific values include Fairness, Compassion, Environmental sustainability, and Democracy.

Table 3: Overlap between human-specific and LLM-specific unexplained value sets.

Metric	Value
Human unique values	27
LLM unique values	76
Shared values	14
Human-only values	13
LLM-only values	62
Jaccard similarity	0.157

4.4 Value Diversity Is a Smooth Gradient, Not a Cherry-Picked Effect

The comparison of extreme exemplars (Section 4.3.1) could be accused of cherry-picking. To rule this out, we generated Kaleido value expressions for all 10,826 human rationales and sorted them by reconstruction error. Figure 6 shows the result: value diversity increases smoothly and monotonically with reconstruction error. Non-overlapping bins of 200 rationales range from 25–31 unique values at the lowest errors to 49–60 unique values at the highest (Pearson $r = 0.393$, $p = 0.003$). The sliding window analysis ($w = 200$, step $= 50$) yields $r = 0.935$ ($p < 0.0001$), confirming the relationship is continuous across the full error range. The "missing dimensions" of LLM moral reasoning encode systematically rarer and more diverse values, not a handful of outliers.

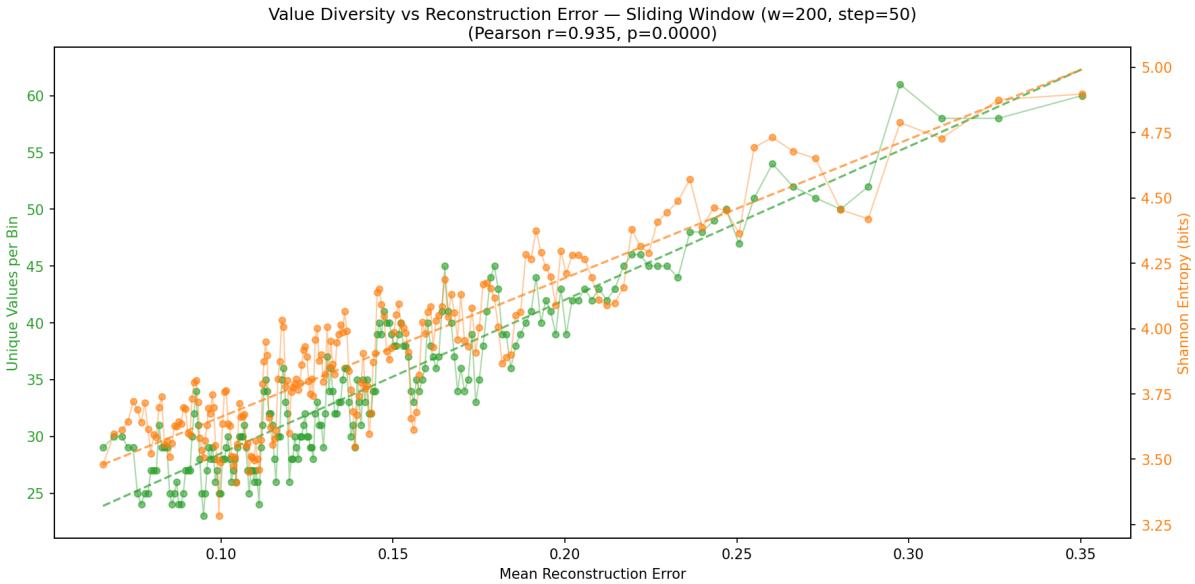

Figure 6: Value diversity gradient across all 10,826 human rationales (sliding window, $w = 200$, step $= 50$). Green: unique values per bin (left axis). Orange: Shannon entropy (right axis). Both increase monotonically with reconstruction error.

4.5 The Gap Widens With Consensus

After controlling for sample size via subsampling (Table 4), the human dimensionality advantage exists at every consensus level but grows with higher moral consensus:

Table 4: Subsampled components for 90% variance by consensus bucket. Each non-human source is subsampled to match human n per bucket (5 seeds, $\pm$ std).

Consensus	n (matched)	Human	All-LLM	Gap
Low (< 0.5)	220	115	95 ± 0.4	+20
Medium (0.5–0.8)	902	259	204 ± 0.7	+55
High (> 0.8)	9,701	444	325 ± 0.9	+119

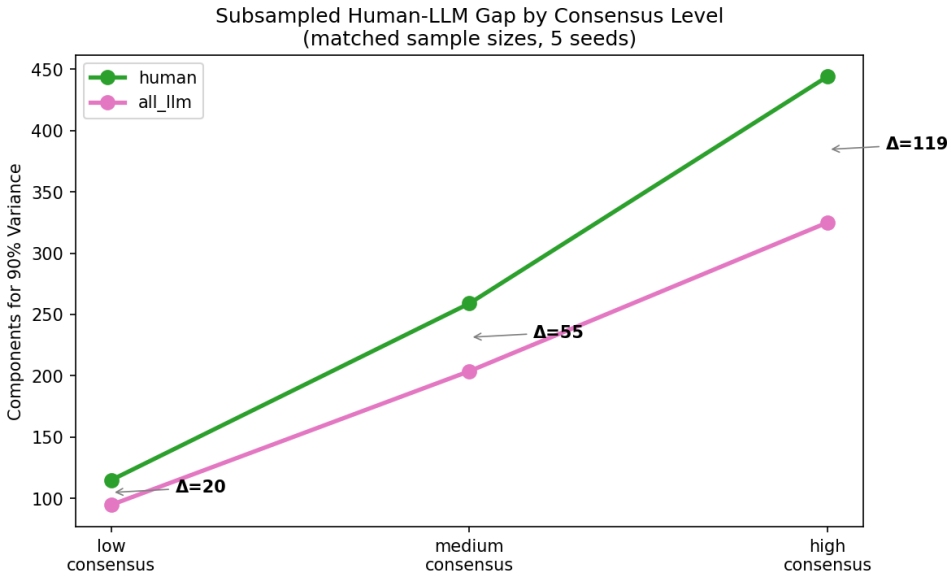

Figure 7: Components for 90% variance by consensus level (subsampled). Shaded bands show ± 1 std across seeds. The gap widens from +20 to +119 as consensus increases.

The original uncontrolled analysis produced a spurious reversal (human 115 vs. all-LLM 256 in the low bucket) because components for 90% is bounded by $\min(n, d)$, and the low bucket had 220 human embeddings vs. 4,399 LLM embeddings. Subsampling corrects this entirely.

4.6 LLMs Do Not Diversify on Controversial Dilemmas

Per-dilemma LLM cosine distance correlates negligibly with community agreement (Pearson $r = -0.059$, Spearman $\rho = -0.062$; both $p < 0.0001$, $n = 10{,}823$). Despite statistical significance (due to large n), the effect size is negligible: LLMs produce the same degree of inter-model convergence whether a dilemma is morally clear-cut or genuinely contested (Figure 8).

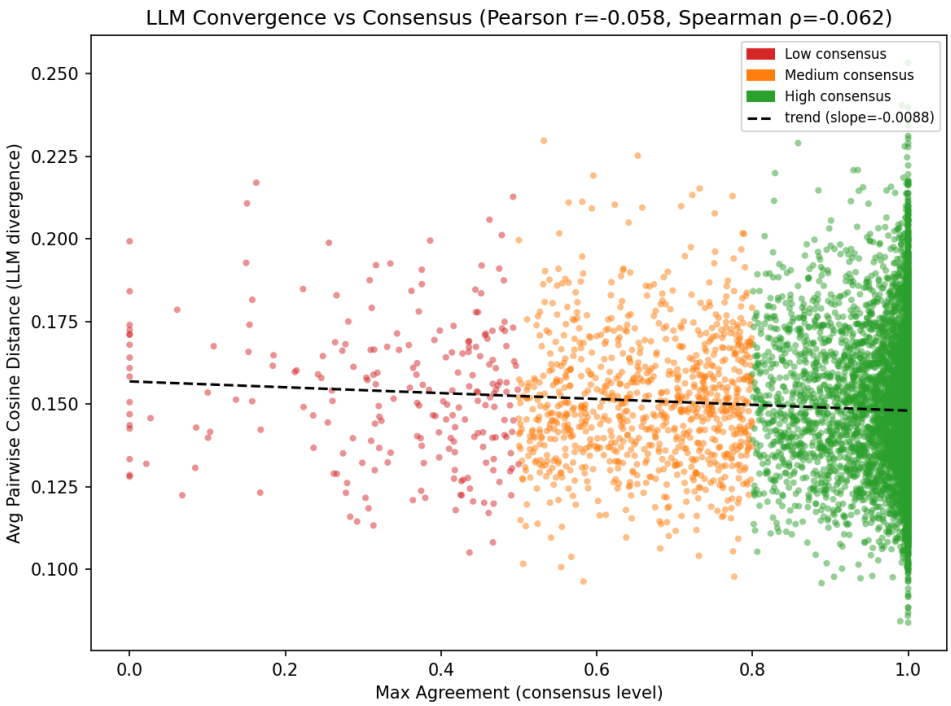

Figure 8: Average pairwise cosine distance among LLM embeddings per dilemma vs. max community agreement. A diverse moral reasoning panel should show higher distance (more disagreement) on controversial dilemmas; LLMs do not.

5 Interpretation

5.1 The Core Finding: Plurality Collapse Is Real but Nuanced

LLMs occupy a measurable subspace of human moral reasoning. The evidence is convergent: lower dimensionality, asymmetric cross-projection, and narrower value expression vocabularies all point in the same direction. However, the effect is not total—LLMs still capture 81.6% of human variance, and the subspace they occupy overlaps substantially with the most common human moral framings (Respect, Family harmony, Autonomy).

The collapse is concentrated in the "long tail" of moral reasoning: rare values, culturally specific perspectives, and situational concreteness that humans bring to moral discourse but that LLMs smooth over in favor of generic assessments. This is precisely the kind of diversity that matters most for representing minority viewpoints in moral deliberation.

Notably, this finding converges with Russo et al.'s count-based result that 81.6% of LLM moral value expressions concentrate in the top 10 value categories, compared to only 35.2% for humans. Our embedding-geometry approach and their frequency-distribution approach are methodologically independent—different data representations, different metrics, different analysis pipelines—yet arrive at the same conclusion: LLMs systematically compress the diversity of moral reasoning. This methodological triangulation significantly strengthens both findings.

5.2 The Stylistic Confound

Approximately 30% of the top distinguishing TF-IDF terms for poorly-captured human rationales are stylistic markers (Reddit slang, profanity, abbreviations) rather than moral concepts. The Kaleido encoder treats register as meaningful signal, so the 18.4% unexplained variance figure includes both genuine moral diversity and stylistic differences. If the stylistic component contributed zero moral signal, the purely content-driven collapse would be approximately

$0.70 \times 18.4\% \approx 12.9\%$. However, this lower bound likely understates the true figure: some "stylistic" terms carry moral meaning—profanity can signal moral outrage, and "nta" is itself a moral verdict. The honest range is **12.9%–18.4%** of human variance, depending on how much moral content the stylistic terms carry. The 2.1× difference in unique value expressions between poorly-captured and well-captured human rationales confirms that the effect is not purely stylistic regardless of where in this range the true value falls.

5.3 High-Consensus Dilemmas: Where Collapse Matters Most

Counterintuitively, the dimensionality gap is largest on dilemmas where humans agree on the verdict ($\Delta = +119$ components) and smallest on controversial ones ($\Delta = +20$). This suggests that plurality collapse is not about LLMs struggling with moral ambiguity—it is about LLMs defaulting to generic reasoning on "easy" cases where humans express rich, diverse perspectives. On clear-cut dilemmas, humans converge on the verdict but diverge in their reasoning; LLMs converge on both.

5.4 LLM Convergence as a Signature of Collapse

The near-zero correlation between LLM inter-model distance and consensus level ($r = -0.059$) provides dilemma-level evidence of collapse. A genuinely diverse panel of moral reasoners should show more disagreement on contested cases. The seven LLMs studied here, despite different architectures and training data, maintain essentially constant inter-model agreement across the full spectrum of moral controversy. This uniformity is consistent with shared training biases (common pretraining corpora, RLHF alignment toward similar value profiles) producing convergent moral representations regardless of the specific model.

5.5 Limitations

Several caveats bound our conclusions:

1. The 30% stylistic component inflates the apparent moral diversity gap.
2. Each dilemma has one human rationale (the top Reddit comment), while LLMs provide 2–3, potentially underestimating within-dilemma human diversity.
3. Kaleido-XL was trained on GPT-4 outputs (via ValuePrism), potentially biasing the embedding space toward GPT-4's moral ontology and underrepresenting human-specific moral concepts.
4. The cross-projection asymmetry uses different k values (448 vs. 347), creating a mechanical advantage for the human-to-LLM direction, though the asymmetry holds at matched k.
5. The low-consensus bucket contains only 220 dilemmas, limiting statistical power for that stratum.
6. All findings are specific to AITA-style everyday moral dilemmas and may not generalize to other moral reasoning contexts.

5.6 Implications

If LLMs are deployed as moral reasoning aids, content moderators, or ethical advisors, their outputs will systematically underrepresent the diversity of human moral perspectives. This underrepresentation is not random—it specifically affects rare values, culturally grounded reasoning, and the situational specificity that characterizes real human moral deliberation. Efforts to improve LLM moral reasoning diversity should focus not on the dominant moral framings (which LLMs already capture well) but on the long tail of moral values that define genuine moral plurality.